Factors Influencing Performance, Motivation and Stress Levels of Software Developers in Remote Work: A Leadership Role Analysis

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Abstract

The transition to remote work has fundamentally transformed workplace dynamics in the software development industry, yet quantitative research examining leadership role differences remains limited in Iberoamerican countries. While developed nations have conducted extensive statistical analyses on remote work impacts, Latin American regions including Argentina, Colombia, Spain, and Mexico are significantly underrepresented in this research domain. This study addresses this critical gap by conducting a quantitative analysis of how managerial and individual contributor roles experience remote work differently across these four Iberoamerican countries. We collected survey data from 232 software professionals and applied statistical methods including t-tests, ANOVA, and effect size calculations to investigate differences in performance, motivation, and stress perceptions between leadership roles across demographic variables including country, gender, and remote work adoption level. Our findings reveal substantial role-based disparities in remote work outcomes. Managers derive significantly greater benefits from remote work compared to individual contributors, demonstrating large effect sizes in team milestones (Cohen's d = 1.34, p = 0.0005), motivation (d = 1.30, p ¡ 0.0001), and problem-solving abilities (d = 1.30, p = 0.0013), while experiencing significant reductions in coordination issues (d = -1.04, p = 0.0053) and communication problems (d = -0.94, p = 0.0003). Individual contributors show more modest benefits, primarily in personal well-being areas such as comfort (d = 0.59, p = 0.0030) and autonomy (d = 0.39, p = 0.0115). Gender analysis reveals pronounced disparities, with male managers experiencing the strongest positive effects across 23 metrics, while female individual contributors face unique challenges including decreased team productivity. These empirical findings provide the first comprehensive quantitative evidence of role-based remote work differences in the Iberoamerican context, suggesting that organizations should implement differentiated remote work policies tailored to specific leadership roles and demographic groups, with particular attention to addressing gender disparities and supporting individual contributors' coordination needs.

Keywords: Remote Work, Software Development, Leadership Roles, Iberoamerican Countries, Motivation, Stress, Performance, Quantitative Analysis

1 Introduction

In recent years, remote work has become a widely adopted practice, especially in the technology sector across Iberoamerican countries. While some developed countries have extensively studied remote work impacts through both descriptive and statistical analyses, analyses of the Iberoamerican region have predominantly been descriptive, with relatively limited emphasis on qualitative approaches. This gap is particularly concerning given the unique cultural, economic, and organizational contexts that characterize software development in these region, which may influence how remote work affects developer experiences differently than in developed countries contexts.

Software developers and similar professionals have experienced a significant shift in their work environment, transitioning from traditional office-based settings to remote or hybrid work models. This shift has brought multiple benefits, such as increased flexibility and reduced commuting times. However, it has also introduced new challenges, particularly regarding workers' motivation and stress levels. Understanding these impacts requires examining not only performance metrics but also motivation and stress, as productivity cannot be reduced to a single dimension but rather represents a set of interconnected metrics that include employee well-being and psychological factors [1].

The central challenge that emerged from our preliminary analysis is understanding how different professional roles—specifically managerial versus individual contributor positions—experience remote work differently across various demographic dimensions. This problem has taken particular relevance because our data reveals significant disparities in how these two groups perceive and benefit from remote work arrangements, with variations further influenced by country, gender, and organizational factors.

For the purposes of this study, we define managerial roles as positions involving people management responsibilities, including supervising team members, making personnel decisions, and having accountability for team outcomes. Individual contributors, in contrast, are professionals who contribute primarily through their individual technical work without direct people management responsibilities [2].

Our investigation focuses on understanding how these role-based differences manifest across different levels of remote work adoption (office-first vs. remote-first approaches) and gender groups (male/female). This analysis is key because the hierarchical organizational structures common in Latin American companies, combined with cultural factors around power distance and leadership dynamics, may influence remote work experiences differently than in other regions.

The unique responsibilities, communication patterns, and decision-making authority of managers may create distinct challenges and opportunities in remote settings compared to individual contributors. However, these differences may be amplified or moderated by demographic factors such as gender, creating complex interaction patterns that require systematic investigation to understand and address effectively.

This study seeks to answer the following research question: *How do perceptions of performance, stress, and motivation differ between managerial and individual contributor roles in the post-pandemic work environment across Iberoamerican countries, and how do these differences vary by gender, and remote work level?*

This question focuses specifically on understanding the differential impacts of remote work on managers versus individual contributors while examining how demographic variables moderate these differences, aiming to provide a better understanding of remote work experiences within the Iberoamerican context. In this way, this research is willing to identify and analyze how managerial and individual contributor roles experience remote work differently in terms of performance, motivation, and stress levels across Iberoamerican countries, with particular attention to demographic variations in these patterns. By leveraging quantitative analysis methods, this study offers a perspective that has been under studied from previous research in this regional context.

1.1 Contributions

This study aims to make three key contributions. First, it applies statistical methods to analyze role-based differences in remote work experiences within the Iberoamerican context, where few quantitative studies exist. Unlike developed countries, which have extensive statistical research on remote work impacts, Latin American countries have minimal representation in the academic literature, making this study particularly valuable for understanding regional differences in leadership role experiences. Second, it provides a general analysis of how demographic factors (gender, remote work level) interact with leadership roles to influence remote work outcomes, offering insights into the complex patterns that shape remote work experiences across different population groups. Third, it offers recommendations for developing role-specific and demographically-sensitive remote work strategies that account for the unique organizational and cultural characteristics of Iberoamerican contexts, adopting more flexible and customized approaches to remote work policy.

1.2 Document Structure

The paper is structured as follows: Section 2 elaborates on the theoretical background and existing literature on remote work impacts, with particular attention to the gap in Iberoamerican research and leadership role differences. Section 3 describes the quantitative methodology employed to analyze data from 232 software professionals across Argentina, Colombia, Spain, and Mexico, focusing on role-based comparisons. Section 4 presents the statistical analysis results, emphasizing differences between managerial and individual contributor roles across demographic variables. Section 5 provides recommendations and identifies follow-up steps for organizations and future research. Section 6 contains the conclusions, and the annexes provide detailed statistical tables and supplementary analysis.

2 Related Literature

2.1 Theoretical Background

The shift to remote work has changed workplace dynamics, affecting employee motivation and stress levels. Various psychological theories help explain how these factors interact in a remote work setting, particularly for software developers. This section reviews two relevant theoretical perspectives:

The Self-Determination Theory, developed by Ryan and Deci [3], identifies three fundamental human needs that drive motivation and well-being: autonomy, competence, and relatedness. In remote work contexts, these needs may be fulfilled differently depending on one's organizational role. Managers may experience enhanced autonomy through reduced interruptions and greater control over their schedules, while individual contributors might face challenges in relatedness due to reduced informal interactions with peers. A study on software engineers working remotely found that competence and autonomy were positively correlated with job satisfaction, but lower relatedness led to feelings of isolation [4].

A second relevant approach is the hedonic and eudaimonic framework. This perspective distinguishes between two aspects of well-being: hedonic well-being, which relates to short-term pleasure and stress avoidance, and eudaimonic well-being, which is associated with meaning, purpose, and long-term fulfillment [5]. For managers, remote work may enhance eudaimonic well-being through better work-life integration and strategic focus, while individual contributors may experience mixed effects depending on their ability to find meaning and connection in distributed work environments.

These theoretical models offer complementary perspectives on how remote work impacts performance, motivation and stress. Additionally, these models might suggest that remote work impacts this three variables differently based on organizational role, providing a foundation for understanding the managerial versus individual contributor differences proposed in our analysis.

2.2 Studies on Remote Work and Role-Based Differences for Developers

Empirical research has explored how remote work impacts different organizational roles, though most studies have focused on developed countries' contexts. Ford et al. [2] conducted one of the studies on role-based differences, finding not statistically differences between people managers and individual contributors in terms of productivity, opening the possibility of understanding differences between different groups of developers according to their role.

Vizcaíno et al. [5] investigated the psychological effects of remote work on software engineers, finding that stress and motivation significantly influence performance, though their analysis did not specifically examine role-based differences. The study used Partial Least Squares Structural Equation Modeling (PLS-SEM) to establish relationships between these factors and revealed that communication and trust among team members are critical in reducing stress levels—factors that may be experienced differently by managers and individual contributors.

Another study by Santos et al. [6] focused on hybrid work models and their impact on developers' satisfaction. The study concluded that hybrid models allow employees to maintain work-life balance while also benefiting from in-person collaboration when needed. However, the research also identified challenges such as feelings of isolation and difficulty in setting clear work boundaries, which contribute to stress.

Lastly, different methodological approaches have been used to study well-being and stress in remote work environments not focusing on the differences between roles:

- **Survey-Based Research:** Multiple surveys have been conducted to measure employee perceptions of stress, well-being, and performance, with the results primarily analyzed with descriptive methods. [?]
- **Partial Least Squares Structural Equation Modeling:** PLS-SEM has been increasingly used to analyze the causal relationships between well-being factors,

demonstrating their suitability for quantitative analysis in the developers' environments [7] [8]

2.3 Research Gaps

Despite growing interest in remote work research, several critical gaps remain:

2.3.1 Limited Research in Iberoamerican Contexts

Most studies on remote work rely on data from Nordic countries, North America, and other developed regions. Latin American countries have minimal representation in quantitative remote work research. The few existing studies from this region provide valuable but not exhaustive quantitative insights and do not examine role-based differences. This gap is significant because hierarchical organizational structures and cultural factors in Iberoamerican countries may influence managerial and individual contributor experiences differently than in developed countries' contexts.

2.3.2 Limited Research on Leadership Role Differences

While some studies acknowledge that managers and individual contributors may experience remote work differently, few have systematically examined these differences across multiple dimensions. The unique challenges and opportunities associated with leadership roles in remote settings remain largely unexplored, particularly in understanding how managerial responsibilities interact with factors such as gender, and remote work level.

2.3.3 Focus on General Outcomes Rather Than Role-Specific Experiences

Most existing studies examine remote work impacts on general employee populations rather than analyzing how different organizational roles experience specific outcomes. Understanding how managerial versus individual contributor roles differently experience motivation, stress, and performance is critical for developing targeted remote work strategies, especially in regions where such research is limited.

2.3.4 Focus on Performance Over Well-Being

Most existing studies focused on employee performance rather than focusing on motivation and stress perception. While performance metrics are essential, understanding how remote work affects employee well-being and long-term job satisfaction is equally critical for organizations adapting to new work models.

This literature review highlights the critical need for research examining role-based differences in remote work experiences within the Iberoamerican context. Addressing these gaps will contribute to a more comprehensive understanding of how organizational hierarchy interacts with demographic factors to shape remote work outcomes across different cultural and regional contexts.

3 Methodology

This study employs a quantitative approach to analyze how managerial and individual contributor roles differently experience motivation, stress, and performance in hybrid/remote work environments. A structured survey was conducted across four Iberoamerican countries: Argentina, Colombia, Spain, and Mexico. The survey gathered data on participants' perceptions of stress levels, motivation and performance, comparing their experiences in remote and in-person work modalities.

3.1 Research Design and Survey Data Collection

This research follows a cross-sectional survey design, using responses from professionals who have worked in both in-person and remote environments for at least two years. The survey was designed and collected in a previous study by Vizcaíno and Garcés [9]. The survey consists of 100 questions designed to capture participants' perceptions of stress levels, motivation, and performance across different work modalities.

The survey development involved various stages to ensure comprehensive coverage of remote work experiences. The questionnaire initially consisted of questions regarding participants' socio-demographic information and previous experience (age, country of origin, remote work experience, days worked from home per week), followed by sections focusing on their perceptions of productivity, knowledge sharing, motivation, stress levels, and performance metrics. The questions were designed based on previous studies and surveys that addressed similar topics [2] and a focus group conducted with 33 software industry workers from various organizational levels, including both managers and individual contributors. This mixed-methods foundation ensured that the survey captured the specific experiences relevant to different leadership roles in remote work environments. The focus group insights were particularly valuable in identifying role-specific challenges and benefits that might not be apparent from literature review alone.

To classify participants into managerial and individual contributor roles, we used a robust approach that combined self-reported role descriptions with expert validation. Participants were asked directly about their current job title and role within their organization. Given the complexity of role classification in the software industry, we found that established classification systems like the Standard Occupational Classification (SOC) had limitations for our context, as they often categorize technical leadership roles (such as Technical Lead or Team Lead) as individual contributors rather than management positions, which does not align with industry practice where these roles typically involve supervisory responsibilities.

To address this challenge, we conducted a comprehensive role classification process that involved:

1. analyzing self-reported job titles and role descriptions from survey participants,
2. consulting with software industry experts from both Latin American and Spanish contexts to validate role classifications, and
3. applying principles from established research frameworks [2] that distinguish between people management and individual contribution roles.

Through this process, participants who indicated job titles or responsibilities involving team leadership, supervision of other developers, project management oversight, or decision-making authority over technical teams were classified as managers. Participants whose roles focused primarily on individual technical work, coding, and individual project contributions without supervisory responsibilities were classified as individual contributors. This classification approach ensures clear distinction between the two role categories for comparative analysis while accounting for the nuanced reality of software industry organizational structures.

The survey that Vizcaíno and Garcés [9] collected was sent to the first participants on September 26, 2024, and they were given until October 31 of the same year to respond. The objective was to attain 100 responses per country to ensure sufficient statistical power for role-based comparisons across demographic groups, with Mexico being the only country to exceed that number. Despite not meeting the initial goal for all countries, the number of responses obtained from the other countries was considered sufficient to analyze trends and patterns in role-based differences.

Detailed data cleaning procedures were implemented to ensure validity for the analysis. Some responses were received from people who had not worked onsite for at least 2 years, despite this requirement being stated in the survey header, so these were discarded to maintain the integrity of the comparisons. Additionally, incomplete surveys lacking sufficient data for role classification or key outcome measures were removed. The final dataset included 232 responses with clear role classifications: 57 managers (24.8%) and 175 individual contributors (75.2%), distributed across Argentina (36), Colombia (42), Spain (56), and Mexico (98).

3.2 Variables and Measurement

The study examines how demographic and organizational factors interact with leadership roles to influence performance, motivation, and stress levels, with particular emphasis on role-based comparisons.

3.2.1 Independent Variables

- **Leadership Role (Primary Focus):** Managerial (1) vs. Individual Contributors (0)
- **Remote Work Level:** Following Šmite et al. [10] terminology, we classified participants as either "office-first" (low remote work, primarily office-based with occasional remote work) or "remote-first" (high remote work, primarily remote-based with occasional office visits)
- **Gender:** Male vs. Female

3.2.2 Dependent Variables

The dependent variables were organized into three main categories based on established frameworks from previous qualitative research conducted with Colombian software developers through focus groups in Bogotá and Medellín [11]:

Performance metrics include: Performance, Team Productivity, Team Milestones, Concentration, Knowledge Exchange, Communication with Peers, Communication with Boss, Employee Engagement, Leadership Problems, Agile Methodology Effectiveness, Task Assignment, Confidence Between Team Members, Collaboration Between Team Members, Qualified Personnel, Project Management Problems, Confidence to Ask Questions, Knowledge About Role, and Shared Leadership.

Stress metrics include: Stress Level, Frustration, Task Pressure, Coordination Issues, Confidence Level, Flexibility, Communication Issues, Team Trust, Time Management, Amount of Hours Worked, and Anger Reactions.

Motivation metrics include: Motivation, Comfort, Trust in Superior, Problem Solving, Work Enjoyment, Work Interest, Company Appreciation, Work-Life Balance, Feedback, Boredom, Attention to Others, Job Loss Fear, Work Environment, Team Influence, Personal Relationships, Autonomy, Conflict Relationships, Belonging, and Creativity.

These metric definitions were derived from the comprehensive factor analysis conducted by Castrillon & Acevedo [11], who identified 83 distinct factors affecting software developer experiences across different work modalities through extensive qualitative research with Colombian professionals. Complete definitions for all dependent variables are provided in Appendix **??**, ensuring that our quantitative measures align with the experiences and terminology used by developers in the Iberoamerican context.

3.3 Data Analysis Approach

The data analysis focuses primarily on role based comparisons while examining how demographic variables affect these differences: The data analysis includes multiple complementary methods:

- **T-tests:** Comparing means between managerial and individual contributor roles for all variables.
- **Remote Work Impact by Role:** Analyzing how high vs. low remote work affects each role differently using separate t-tests for managers and individual contributors.
- **Demographic Interaction Analysis:** Examining how country, gender, and other variables interact with leadership roles to influence outcomes.
- **ANOVA for Multi-Group Comparisons:** Analyzing role differences across multiple demographic categories simultaneously.
- **Effect Size Calculations:** Determining practical significance of role-based differences using Cohen's d.

The analysis was conducted using Python with specialized libraries for statistical analysis, including Pandas, NumPy, SciPy, and Seaborn. For each demographic group, we performed separate analyses of role-based differences to identify group-specific patterns, with particular emphasis on understanding how managerial versus individual contributor experiences vary across genders, and remote work levels.

4 Results

4.1 Role Based Differences in Remote Work Experience

The analysis focuses on how managerial and individual contributor roles experience remote work differently across demographic groups in four Iberoamerican countries. The data reveals significant role-based disparities in remote work outcomes, with distinct patterns emerging across demographic variables.

4.1.1 Demographic Distribution by Leadership Role

The sample of 232 participants demonstrates good representation for role-based analysis. Leadership roles include 57 managers (24.8%) and 175 individual contributors (75.2%), reflecting typical organizational structures in the software industry. This distribution provides sufficient power for statistical comparisons while representing realistic organizational hierarchies.

Remote work level distribution shows 194 participants (83.6%) in high remote work and 38 participants (16.4%) in low remote work. Country distribution includes Argentina (15.5%), Colombia (18.1%), Spain (24.1%), and Mexico (42.2%). Gender distribution shows 184 males (79.3%) and 47 females (20.2%), with role-gender combinations providing adequate representation for interaction analysis except for female managers, where sample size was insufficient for reliable statistical inference.

4.1.2 Overall Metrics by Leadership Role

Initial analysis revealed significant differences in how managers and individual contributors experience their work environment, independent of remote work level. The analysis across performance, stress, and motivation metrics shows distinct patterns between the two role groups.

Performance Metrics: Only one significant difference emerged in performance metrics between managers and individual contributors. Managers reported significantly higher scores in Personal Relationships Performance (3.67 vs. 3.34, p=0.037), suggesting that managers perceive their interpersonal work relationships as more effective than individual contributors do. Other performance metrics, including Team Productivity, Team Milestones, and overall Performance, showed no significant differences between roles.

Stress Metrics: The stress category revealed the most pronounced role-based differences. Managers reported significantly higher Communication Issues (3.44 vs. 3.02, p=0.003) and Stress Level (2.61 vs. 2.25, p=0.047) compared to individual contributors. Conversely, individual contributors showed significantly higher Confidence Level (3.61 vs. 3.35, p=0.038). These findings suggest that managerial roles inherently involve more communication-related challenges and stress, while individual contributors maintain higher confidence levels in their work capabilities.

Motivation Metrics: In motivation-related factors, individual contributors reported significantly better Work Environment perceptions (3.40 vs. 3.09, p=0.032) compared to managers. All other motivation metrics, including overall Motivation, Work Enjoyment, and Autonomy, showed no significant differences between roles, indicating that baseline motivation levels are similar across leadership positions.

Table 1 summarizes the significant baseline differences between managers and individual contributors.

These baseline differences establish that managers face inherently higher stress and communication challenges while maintaining better interpersonal performance relationships. Individual contributors, conversely, report higher confidence levels and better work environment perceptions. These fundamental role-based differences provide important context for understanding how remote work affects each group differently, as explored in subsequent analyses.

Table 1: Significant baseline differences between Managers and Individual Contributors

Category	Significant Differences	Direction
Performance	Personal Relationships Performance	Managers > Individual Contributors
Stress	Communication Issues	Managers > Individual Contributors
Stress	Stress Level	Managers > Individual Contributors
Stress	Confidence Level	Individual Contributors > Managers
Motivation	Work Environment	Individual Contributors > Managers

4.2 Remote Work Impact by Leadership Role

The most significant finding of our study is that remote work affects managerial and individual contributor roles dramatically differently, with managers deriving substantially greater benefits across multiple dimensions.

4.2.1 Remote Work Effects on Managers

Managers experience dramatic benefits from remote work across multiple dimensions. The most significant positive effects include:

Major Positive Effects for Managers in Remote-First Environments:

- **Team Milestones:** +1.34 points (p=0.0005) - Large effect indicating substantially better milestone achievement
- **Motivation:** +1.30 points (p¡0.0001) - Large effect showing enhanced personal motivation
- **Problem Solving:** +1.30 points (p=0.0013) - Large effect demonstrating improved problem-solving capabilities
- **Team Productivity:** +1.23 points (p=0.0135) - Large effect reflecting better team coordination
- **Feedback:** +1.31 points (p¡0.0001) - Large effect indicating improved feedback processes

Major Reductions in Workplace Problems:

- **Coordination Issues:** -1.04 points (p=0.0053) - Large reduction in coordination difficulties
- **Communication Issues:** -0.94 points (p=0.0003) - Large reduction in communication problems
- **Frustration:** -0.85 points (p=0.0167) - Large reduction in workplace frustration
- **Project Management Problems:** -0.83 points (p=0.0020) - Large reduction in project management difficulties

4.2.2 Remote Work Effects on Individual Contributors

Individual contributors benefit from remote work, but their gains are more modest and focused on personal well-being rather than team outcomes:

Positive Effects for Individual Contributors in Remote-First Environments:

- **Comfort:** +0.59 points (p=0.0030) - Moderate increase in workplace comfort
- **Autonomy:** +0.39 points (p=0.0115) - Small but significant increase in autonomy
- **Performance:** +0.37 points (p=0.0256) - Small improvement in individual performance
- **Creativity:** +0.34 points (p=0.0366) - Small enhancement in creative work
- **Confidence Level:** +0.34 points (p=0.0297) - Small increase in confidence

Challenges for Individual Contributors:

- **Conflict Relationships:** -0.29 points (p=0.049) - Small but significant increase in relationship conflicts

4.2.3 Role-Based Comparative Analysis

The analysis reveals an interesting pattern: managers experience remote work benefits that are 3-4 times larger in magnitude than those experienced by individual contributors. For managers, remote work creates substantial improvements in team coordination, motivation, and productivity metrics, while simultaneously reducing workplace problems and frustrations. Individual contributors, while still benefiting from remote work, experience much more modest gains focused primarily on personal well-being rather than team or organizational outcomes.

Critical Differences:

1. **Leadership Problems:** Individual contributors report significantly more leadership problems in remote environments (+0.30), while managers report fewer leadership problems (-0.67), suggesting managers find remote leadership easier while individual contributors feel less supported.
2. **Team Coordination:** Managers report dramatically fewer coordination issues (-1.04) in remote work, while individual contributors show no significant improvement, indicating managers may be better equipped to handle remote coordination challenges.
3. **Motivation and Performance:** Managers show large gains in motivation (+1.30) and team productivity (+1.23), while individual contributors show smaller performance gains (+0.37) and non-significant motivation changes.

4.3 Demographic Interactions with Leadership Roles

4.3.1 Gender Analysis

Our gender analysis reveals significant variations in how remote work affects males and females in different leadership positions. This breakdown provides valuable insights into how gender and leadership roles interact to influence remote work experiences.

Table 2: Distribution of Participants by Gender and Leadership Role

Gender	Managers	Individual contributors	Total
Male	42	142	184
Female	15	32	47
Total	57	174	231

Table 2 shows the distribution of participants by gender and leadership role.

The sample includes predominantly male participants (79.7%), with gender disparities more pronounced in management positions where males constitute 73.7% of managers compared to females at 26.3%. This distribution reflects typical gender imbalances in technical leadership roles in the software industry.

Table 3 shows the performance and stress metric differences by gender and leadership role.

Table 3: Performance and Stress Metric Differences (High Remote - Low Remote) by Gender and Leadership Role

Metric	Male Managers	Male Individual contributors	Female Managers	Female Individual contributors
Performance	+0.97*	+0.48*	Insufficient data	+0.13
Team Productivity	+1.47*	+0.40*	Insufficient data	−0.22
Team Milestones	+1.36*	+0.24	Insufficient data	+0.23
Motivation	+1.25*	+0.35	Insufficient data	+0.34
Work Enjoyment	+1.31*	+0.48*	Insufficient data	+0.04
Creativity	+1.03	+0.41*	Insufficient data	+0.34
Work-Life Balance	+0.61	+0.44	Insufficient data	+0.41
Stress Level	−0.69	−0.11	Insufficient data	+0.19
Confidence Level	+0.81*	+0.44*	Insufficient data	+0.16
Task Pressure	−0.53	−0.21	Insufficient data	−0.13

*Statistically significant at $p<0.05$

Our analysis reveals several interesting patterns in how remote work affects different gender-leadership combinations:

- **Male Managers** experience the strongest positive effects from remote work across nearly all metrics. For this group, remote work is associated with substantial improvements in performance (+0.97 points), team productivity (+1.47 points), and team milestones (+1.36 points). They also report the largest increases in motivation (+1.25 points) and work enjoyment (+1.31 points). These effects are statistically significant and notably stronger than those observed in other groups.

- **Male Individual contributors** show moderate positive effects from remote work, with significant improvements in performance (+0.48 points), team productivity (+0.40 points), work enjoyment (+0.48 points), and creativity (+0.41 points). While beneficial, these effects are consistently smaller in magnitude compared to those for male managers.
- **Female Individual contributors** show mixed but generally positive effects from remote work. Their improvements in performance (+0.13 points), creativity (+0.34 points), and work-life balance (+0.41 points) are modest and not statistically significant. Interestingly, this is the only group showing a negative effect on team productivity (-0.22 points), suggesting potential challenges with team collaboration in remote settings.
- **Female Managers** had insufficient data for reliable statistical analysis, highlighting a gap in our understanding of how remote work affects women in leadership positions.

4.3.2 Country Analysis

Role-based differences manifest differently across Iberoamerican countries:

Spain: Managers show particularly strong performance benefits (+0.50 points significant effect), while individual contributors show more modest gains.

Mexico: Both roles benefit from remote work, but managers show stronger motivation improvements (+0.80 points significant for overall sample, with managers driving this effect).

Argentina and Colombia: Limited sample sizes within role categories prevent definitive conclusions, but trends suggest similar patterns of stronger managerial benefits.

4.4 Summary of Key Findings

Our comprehensive analysis of remote work's impact on software developers reveals several important patterns:

- **Overall Positive Impact:** Remote work is generally associated with positive outcomes across multiple metrics, with high remote work participants reporting better performance, motivation, work enjoyment, and creativity compared to low remote work participants.
- **Leadership Role Differences:** Managers and individual contributors experience remote work differently:
  - Managers experience more substantial benefits in team productivity, team milestones, and motivation
  - Managers also report reduced frustration and task pressure in high remote work settings
  - Individual contributors see more modest benefits in performance and creativity
  - Coordination issues show opposite effects: managers report fewer issues with high remote work, while individual contributors report the same or more
- **Country Differences:**

  - Spain shows significant improvements in performance with high remote work, especially for Managerial roles
  - Mexico shows significant improvements in motivation for both Managerial and individual contributor's roles

- **Gender Differences:**
  - Males show significant benefits across 23 metrics, while females show no statistically significant effects
  - Male managers receive the strongest benefits from remote work, with dramatic improvements in performance (+0.97), team productivity (+1.47), and motivation (+1.25)
  - Female individual contributors report decreased team productivity (-0.22) in remote settings, contrasting with males' positive experience
  - Work-life balance improvements are stronger for male managers (+0.61) than for other groups
  - Remote work appears to particularly benefit males in leadership positions, suggesting potential gender disparities in remote work experiences

- **Gender-Leadership Interactions:**
  - Leadership role amplifies remote work benefits for males, with male managers showing effects 2-3 times stronger than male individual contributors
  - Male managers report the largest reductions in stress level (-0.69) and task pressure (-0.53), indicating remote work particularly alleviates pressure for men in leadership
  - Female employees show more modest and largely non-significant benefits from remote work compared to their male counterparts
  - The lack of sufficient data for female managers represents an important gap in understanding how women in leadership experience remote work

These findings suggest that remote work generally enhances software developers' well-being and performance, but the magnitude and nature of these benefits vary considerably based on leadership role, country, and gender. The gender-leadership interactions reveal that remote work is not experienced uniformly across demographic combinations, with male managers benefiting most substantially. These variations have important implications for how organizations design and implement remote work policies to ensure equitable benefits across all employee groups.

4.5 Integration with Local Qualitative Research

Our quantitative findings show convergence with qualitative insights from previous research conducted with Colombian software professionals [11]. This convergence strengthens the validity of our role-based findings by providing triangulation between quantitative patterns and developers' reported experiences.

For managers: The large quantitative effects we observed for managers (+1.30 motivation, +1.34 team milestones, -1.04 coordination issues) align with qualitative reports from Colombian focus groups where managers mentioned feeling more empowered and efficient in remote settings. The qualitative data provides context for why these benefits occur, with managers reporting better focus time, reduced meeting interruptions, and enhanced strategic thinking capabilities in remote environments.

For Individual contributors: Our finding that individual contributors report more leadership problems (+0.30 points) and smaller benefits overall corresponds with Colombian focus group participants who mentioned feeling disconnected from leadership and experiencing reduced mentoring opportunities in remote settings. The qualitative research provides important context for understanding why individual contributors face these challenges, including reduced informal interactions and fewer opportunities for spontaneous collaboration.

For both groups: Several variables that showed significant role-based differences in our statistical analysis were also prominent themes in the Colombian qualitative research:

- Coordination challenges for individual contributors were mentioned frequently in focus groups
- Enhanced problem-solving for managers was noted as a recurring theme
- Communication issues varying by role were identified as a central concern
- Motivation differences between hierarchical levels were discussed extensively

5 Recommendations and Follow-Up Steps

Based on our analysis, we offer the following recommendations for organizations implementing or optimizing remote work for software developers:

5.1 Leadership-Specific Recommendations

5.1.1 Create Tailored Remote Work Approaches

Rather than one big policy, develop distinct remote work frameworks that recognize the different ways managers and team members experience remote environments. Managers thrive with more autonomy and decision-making power, while team members need different support structures.

Example implementation: Establish role-based remote work schedules—3-4 remote days/week for managers to leverage their significant productivity gains and 2-3 remote days/week for individual contributors to maintain sufficient in-person coordination opportunities.

5.1.2 Build Better Coordination Systems for Team Members

Our findings show that individual contributors often feel disconnected in remote settings. Implement specific coordination tools and regular check-in practices to help team members stay connected with their colleagues and maintain project alignment.

Example implementation: Deploy digital coordination dashboards like Monday.com or Asana with daily automated check-ins specifically for individual contributors to address their increase in coordination issues in virtual environments.

5.1.3 Develop Remote Leadership Skills

Invest in training that helps managers leverage their strengths in remote environments. Since managers show significant productivity and motivation gains when working remotely, help them develop skills to extend these benefits to their teams through effective remote leadership practices.

Example implementation: Institute virtual training sessions for managers focusing on remote team management skills.

5.1.4 Improve Communication for Managers

While managers benefit greatly from remote work, they still report communication challenges. Create clear communication frameworks that help leaders stay connected with their teams without increasing their stress levels.

Example implementation: Develop a documented communication framework specifying which channels to use for different types of communications, appropriate response times, and escalation paths to address managers' 0.42-point higher communication issues.

5.2 Gender-Focused Recommendations

5.2.1 Address Male-Female Disparities

Our findings reveal that male software professionals particularly managers, are experiencing significantly more benefits from remote work than their female colleagues. Organizations should directly address this gap by examining what factors are enabling male success in remote settings.

Example implementation: Establish gender-balanced remote practices including structured check-ins for female team members to address their -0.22 team productivity challenge and mentoring pairs between male managers (who saw a 1.47-point productivity boost) and female managers.

5.2.2 Support Female Team Members

Female individual contributors report unique challenges with team productivity in remote settings. Create additional support structures and check-in systems to ensure women feel equally connected and productive when working remotely.

Example implementation: Form monthly women's remote work networks specifically for female developers to share strategies, challenges, and solutions, with special attention to team productivity issues where they reported decreased performance.

5.3 Regional Considerations for Iberoamerican Organizations

Latin American organizations often have distinct hierarchical cultures that may amplify the role-based differences observed in our study.

- Provide cultural competency training for remote leadership that acknowledges regional hierarchical expectations
- Create communication protocols that respect cultural norms while promoting effective remote collaboration

- Develop region-specific remote work guidelines that account for local organizational cultures

5.4 Follow-Up Research

5.4.1 Study Long-Term Effects

This research captures a snapshot in time. Following these same teams over months and years would help us understand whether the observed benefits and challenges of remote work persist or evolve.

5.4.2 Female Leadership Research

Address the critical gap in understanding female managers' remote work experiences through targeted data collection and analysis.

5.4.3 Talk to Developers

While our numbers tell an important story, direct conversations with developers would provide deeper insights into why remote work affects different groups so differently. Interviews and focus groups would complement our quantitative findings.

5.4.4 Integration of Qualitative Insights

Review more existing qualitative research, like the work by Castrillón & Acevedo [11] on Colombian focus groups, to identify whether the variables showing significant role-based differences in our quantitative analysis are also mentioned in qualitative reports, providing convergent evidence for the role-based patterns observed.

5.4.5 Test Specific Solutions

Design and evaluate targeted interventions to address the coordination challenges for individual contributors and the potential differences for each company type, country and gender.

5.4.6 Broaden Our Understanding

Extend this analysis to include other important factors like age, parenting status, years of experience, and personality traits, which likely influence how people experience remote work.

5.4.7 Stronger Mathematical Analysis

Implement Partial Least Squares Structural Equation Modeling (PLS-SEM) with instrumental variables to strengthen causal inference beyond our current correlational findings. This more sophisticated mathematical analysis would allow us to identify complex relationships between variables, control confounding factors, and establish stronger evidence for causality in remote work effects.

By implementing these recommendations with concrete action steps, organizations can create remote work environments that benefit all team members while addressing the specific needs and challenges faced by different groups.

6 Conclusion

This study has provided the first comprehensive quantitative analysis of how leadership roles influence remote work experiences in the Iberoamerican software development context. Through statistical analysis of data from 232 software professionals across Argentina, Colombia, Spain, and Mexico, we have uncovered significant role-based disparities that have important implications for remote work policy and practice.

Our findings demonstrate that remote work is not a uniform experience across organizational hierarchies. Managers derive substantially greater benefits from remote work arrangements, showing large effect sizes in critical areas such as team milestones (d = 1.34), motivation (d = 1.30), and problem-solving capabilities (d = 1.30), while simultaneously experiencing significant reductions in coordination and communication challenges. Individual contributors, while still benefiting from remote work, experience more modest gains focused primarily on personal well-being metrics such as comfort and autonomy.

The gender analysis reveals concerning disparities that warrant immediate organizational attention. Male managers experience the strongest positive effects from remote work across multiple dimensions, while female individual contributors face unique challenges, including decreased team productivity. The insufficient representation of female managers in our sample highlights a critical gap in understanding how women in leadership positions experience remote work in the Iberoamerican context.

These findings challenge the one-size-fits-all approach to remote work policy that many organizations have adopted. Our evidence suggests that effective remote work strategies must be differentiated by leadership role and demographic characteristics. Managers may benefit from increased remote work flexibility to leverage their demonstrated productivity gains, while individual contributors may require more structured support systems and coordination mechanisms to address their specific challenges in distributed work environments.

The Iberoamerican context adds additional complexity to these patterns. The hierarchical organizational cultures common in Latin American companies may amplify the role-based differences we observed, making targeted interventions even more critical for ensuring equitable remote work experiences across all employee groups.

From a theoretical perspective, our findings provide empirical support for Self-Determination Theory's predictions about how autonomy, competence, and relatedness needs are fulfilled differently across organizational roles in remote settings. Managers appear to experience enhanced autonomy and competence in remote environments, while individual contributors may face challenges in maintaining relatedness and receiving adequate support.

6.1 Limitations

Several limitations should be acknowledged. First, the cross-sectional design limits our ability to establish causal relationships between remote work and outcomes. Second, the underrepresentation of female managers prevents comprehensive analysis of gender-leadership interactions. Third, the survey-based methodology relies on self-reported perceptions, which may be subject to various biases. Finally, the focus on four Iberoamerican countries, while valuable for regional insights, may limit generalizability to other cultural contexts.

6.2 Future Research Directions

Future research should address these limitations through longitudinal studies that can establish causal relationships and track the evolution of remote work effects over time. Targeted data collection focusing on female managers and other underrepresented groups would provide a more complete picture of demographic interactions with remote work. Mixed-methods approaches combining quantitative analysis with in-depth qualitative research would offer richer insights into the mechanisms underlying the patterns we observed.

Additionally, advanced statistical modeling techniques such as Partial Least Squares Structural Equation Modeling (PLS-SEM) could help identify complex relationships between variables and strengthen causal inference. Cross-cultural studies extending this analysis to other regional contexts would enhance our understanding of how cultural factors influence role-based remote work experiences.

6.3 Practical Implications

For organizations in the Iberoamerican region and beyond, our findings suggest several practical implications. First, remote work policies should be differentiated by leadership role, with managers potentially benefiting from greater remote work flexibility and individual contributors requiring more structured coordination support. Second, organizations must actively address gender disparities in remote work experiences through targeted interventions and support systems. Third, regular assessment of remote work impacts across different demographic groups is essential for ensuring equitable outcomes.

The evidence presented in this study provides a foundation for evidence-based remote work policy development that recognizes the complexity of modern software development organizations. By acknowledging and addressing the differential impacts of remote work across leadership roles and demographic groups, organizations can create more effective and equitable distributed work environments that benefit all team members while maintaining high levels of productivity and employee well-being.

In conclusion, this research contributes to the growing body of knowledge on remote work by providing the first quantitative analysis of leadership role differences in the Iberoamerican context. The substantial disparities we identified underscore the importance of moving beyond universal remote work policies toward more nuanced, role-specific, and demographically-sensitive approaches that can maximize the benefits of distributed work arrangements while addressing the unique challenges faced by different groups within software development organizations.

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A Annexes

A.1 Annex A

A.1.1 Table A.1: Performance Metrics by Leadership Role

Table 4: Performance Metrics by Leadership Role

Metric	**Managers (Mean)**	**Individual contributors (Mean)**	**Difference**	**Significant at α=0.05**
Performance	3.79	3.84	−0.05	No
Team Productivity	3.56	3.70	−0.14	No
Team Milestones	3.40	3.58	−0.18	No
Concentration	3.74	3.59	0.14	No
Knowledge Exchange	2.81	2.90	−0.09	No
Communication With Peers	2.84	2.77	0.08	No
Communication With Boss	2.86	2.89	−0.03	No

A.1.2 Table A.2: Stress Metrics by Leadership Role

Table 5: Stress Metrics by Leadership Role

Metric	**Managers (Mean)**	**Individual contributors (Mean)**	**Difference**	**Significant at α=0.05**
Stress Level	2.61	2.25	0.36	Yes
Confidence Level	3.35	3.61	−0.26	Yes
Frustration	2.77	2.50	0.28	No
Task Pressure	3.07	2.94	0.13	No
Flexibility	4.42	4.22	0.20	No
Communication Issues	3.44	3.02	0.42	Yes
Team Trust	2.77	2.97	−0.19	No
Time Management	3.86	3.78	0.08	No
Coordination Issues	3.11	2.97	0.13	No

A.1.3 Table A.3: Motivation Metrics by Leadership Role

Table 6: Motivation Metrics by Leadership Role

Metric	**Managers (Mean)**	**Individual contributors (Mean)**	**Difference**	**Significant at α=0.05**
Motivation	3.49	3.65	−0.15	No
Comfort	4.40	4.52	−0.12	No
Trust in Superior	3.35	3.52	−0.17	No
Problem Solving	3.61	3.72	−0.11	No
Work Enjoyment	3.79	3.93	−0.14	No
Work Interest	3.32	3.37	−0.06	No
Work Life Balance	3.98	4.10	−0.11	No
Work Environment	3.09	3.40	−0.31	Yes
Creativity	3.54	3.75	−0.20	No

A.2 Annex B: Complete Remote Work Impact Analysis by Leadership Role

A.2.1 Table B.1: Detailed t-test results for managers comparing remote-first vs. office-first conditions

Table 7: Remote Work Impact on Managers: Complete Statistical Results

Metric	**High Remote Mean**	**Low Remote Mean**	**Difference**	**t-statistic**	**p-value**	**Significant**
Team Milestones	3.59	2.25	1.34	4.71	0.0005	Yes
Feedback	3.18	1.88	1.31	6.84	0.0000	Yes
Motivation	3.67	2.38	1.30	5.67	0.0000	Yes
Problem Solving	3.80	2.50	1.30	4.42	0.0013	Yes
Team Productivity	3.73	2.50	1.23	3.03	0.0135	Yes
Confidence ToAsk Questions	3.53	2.38	1.16	3.97	0.0024	Yes
Trust in Superior	3.51	2.38	1.14	3.83	0.0027	Yes
Team Influence	3.27	2.12	1.14	4.29	0.0009	Yes
Work Environment	3.24	2.12	1.12	3.48	0.0061	Yes
Attention to Others	3.10	2.00	1.10	3.70	0.0037	Yes
Lack Empl Engagement	3.18	4.25	−1.07	−3.64	0.0030	Yes
Coordination Issues	2.96	4.00	−1.04	−3.44	0.0053	Yes
Belonging	3.14	2.12	1.02	3.13	0.0104	Yes
Shared Leadership	2.88	3.88	−1.00	−5.82	0.0000	Yes
Communication Issues	3.31	4.25	−0.94	−4.55	0.0003	Yes

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Table 7 continued from previous page

Metric	High Remote Mean	Low Remote Mean	Difference	t-statistic	p-value	Significant
Collab Betw Team-Members	3.00	3.88	−0.88	−4.78	0.0001	Yes
Frustration	2.65	3.50	−0.85	−2.81	0.0167	Yes
ProjectMgmt Problems	2.80	3.62	−0.83	−3.86	0.0020	Yes
Personal Relationships	2.41	1.62	0.78	3.40	0.0035	Yes
Tasks assignment	3.31	2.62	0.68	2.35	0.0400	Yes
Leadership Problems	3.20	3.88	−0.67	−2.74	0.0217	Yes
Knowledge Exchange	2.90	2.25	0.65	3.12	0.0062	Yes
Confidence Betw TeamMembers	3.35	4.00	−0.65	−2.67	0.0155	Yes
Communication With Boss	2.94	2.38	0.56	2.59	0.0217	Yes
Task Pressure	3.00	3.50	−0.50	−2.16	0.0471	Yes
Company Appreciation	3.10	2.62	0.48	2.28	0.0421	Yes
Flexibility	4.47	4.12	0.34	2.14	0.0463	Yes
Work Enjoyment	3.92	3.00	0.92	1.90	0.0926	No
Creativity	3.65	2.88	0.78	2.08	0.0668	No
Qualified Personnel	3.37	2.62	0.74	2.14	0.0603	No
Performance	3.88	3.25	0.63	1.85	0.0953	No
Team Trust	2.86	2.25	0.61	2.20	0.0516	No
Knowledge About-Role	3.04	3.62	−0.58	−1.70	0.1242	No
Confidence Level	3.43	2.88	0.55	2.19	0.0517	No
Communication With Peers	2.92	2.38	0.54	1.87	0.0899	No
Personal Relationships Perf	3.59	4.12	−0.53	−1.40	0.1932	No
Conflict Relationships	2.41	2.88	−0.47	−1.00	0.3433	No
Stress Level	2.55	3.00	−0.45	−1.08	0.3058	No
Concentration	3.80	3.38	0.42	1.22	0.2539	No
Time Management	3.92	3.50	0.42	1.38	0.1936	No
Work Interest	3.37	3.00	0.37	1.01	0.3352	No
Boredom	2.53	2.88	−0.34	−1.28	0.2206	No
Comfort	4.45	4.12	0.32	2.01	0.0592	No
Job Loss Fear	2.94	3.25	−0.31	−1.10	0.2956	No
Autonomy	3.94	3.62	0.31	0.91	0.3866	No

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Table 7 continued from previous page

Metric	High Remote Mean	Low Remote Mean	Difference	t-statistic	p-value	Significant
Anger Reactions	2.65	2.88	−0.22	−0.67	0.5151	No
AgileM Efective-ness	3.55	3.38	0.18	0.44	0.6711	No
Work Life Balance	4.00	3.88	0.12	0.25	0.8107	No

Positive Effects:

- Significantly higher Team Milestones (+1.34 points, $p = 0.0005$)
- Significantly higher Feedback (+1.31 points, $p < 0.0001$)
- Significantly higher Motivation (+1.30 points, $p < 0.0001$)
- Significantly higher Problem Solving (+1.30 points, $p = 0.0013$)
- Significantly higher Team Productivity (+1.23 points, $p = 0.0135$)
- Significantly higher Confidence to Ask Questions (+1.16 points, $p = 0.0024$)
- Significantly higher Trust in Superior (+1.14 points, $p = 0.0027$)
- Significantly higher Team Influence (+1.14 points, $p = 0.0009$)
- Significantly higher Work Environment (+1.12 points, $p = 0.0061$)
- Significantly higher Attention to Others (+1.10 points, $p = 0.0037$)
- Significantly higher Belonging (+1.02 points, $p = 0.0104$)
- Significantly higher Personal Relationships (+0.78 points, $p = 0.0035$)
- Significantly higher Tasks Assignment (+0.68 points, $p = 0.0400$)
- Significantly higher Knowledge Exchange (+0.65 points, $p = 0.0062$)
- Significantly higher Communication with Boss (+0.56 points, $p = 0.0217$)
- Significantly higher Company Appreciation (+0.48 points, $p = 0.0421$)
- Significantly higher Flexibility (+0.34 points, $p = 0.0463$)

Negative Effects:

- Significantly lower Lack of Employee Engagement (−1.07 points, $p = 0.0030$)
- Significantly lower Coordination Issues (−1.04 points, $p = 0.0053$)
- Significantly lower Shared Leadership (−1.00 points, $p < 0.0001$)
- Significantly lower Communication Issues (−0.94 points, $p = 0.0003$)

- Significantly lower Collaboration Between Team Members (-0.88 points, $p = 0.0001$)
- Significantly lower Frustration (-0.85 points, $p = 0.0167$)
- Significantly lower Project Management Problems (-0.83 points, $p = 0.0020$)
- Significantly lower Leadership Problems (-0.67 points, $p = 0.0217$)
- Significantly lower Confidence Between Team Members (-0.65 points, $p = 0.0155$)
- Significantly lower Task Pressure (-0.50 points, $p = 0.0471$)

A.2.2 Table B.2: Detailed t-test results for individual contributors comparing remote-first vs. office-first conditions

Table 8: Remote Work Impact on Individual Contributors: Complete Statistical Results

Metric	High Remote Mean	Low Remote Mean	Difference	t-statistic	p-value	Significant
Comfort	4.62	4.03	0.59	3.19	0.0030	Yes
Flexibility	4.29	3.90	0.39	2.32	0.0256	Yes
Autonomy	3.99	3.60	0.39	2.63	0.0115	Yes
Performance	3.90	3.53	0.37	2.31	0.0256	Yes
Confidence Level	3.67	3.33	0.34	2.24	0.0297	Yes
Creativity	3.81	3.47	0.34	2.16	0.0366	Yes
Problem Solving	3.77	3.47	0.31	2.19	0.0340	Yes
Leadership Problems	3.26	2.97	0.30	2.04	0.0464	Yes
Conflict Relationships	2.21	2.50	−0.29	−2.02	0.0490	Yes
Work Life Balance	4.17	3.77	0.40	1.90	0.0640	No
Work Enjoyment	3.99	3.63	0.36	1.77	0.0842	No
Motivation	3.70	3.40	0.30	1.76	0.0850	No
Coordination Issues	3.01	2.77	0.25	1.83	0.0727	No
Personal Relationships Perf	3.38	3.13	0.25	1.41	0.1650	No
Team Productivity	3.74	3.50	0.24	1.37	0.1785	No
Tasks assignment	3.28	3.03	0.24	1.70	0.0966	No
Qualified Personnel	3.37	3.13	0.23	1.66	0.1047	No
Job Loss Fear	2.96	2.73	0.23	1.34	0.1859	No
Collab Betw TeamMembers	3.12	2.90	0.22	1.77	0.0837	No
Confidence Betw TeamMembers	3.24	3.03	0.21	1.58	0.1199	No
Work Interest	3.41	3.20	0.21	1.15	0.2578	No
Team Trust	2.93	3.13	−0.20	−1.45	0.1540	No
Personal Relationships	2.20	2.40	−0.20	−1.31	0.1958	No
Team Milestones	3.61	3.43	0.18	1.14	0.2586	No
Confidence ToAsk Questions	3.57	3.40	0.17	1.20	0.2372	No
Attention to Others	2.97	3.13	−0.17	−1.17	0.2488	No
Task Pressure	2.91	3.07	−0.16	−1.21	0.2311	No
Work Environment	3.43	3.27	0.16	1.01	0.3181	No
Team Influence	3.19	3.33	−0.15	−1.01	0.3183	No

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Table 8 continued from previous page

Metric	**High Remote Mean**	**Low Remote Mean**	**Difference**	**t-statistic**	**p-value**	**Significant**
Trust in Superior	3.54	3.40	0.14	0.96	0.3397	No
Communication Issues	3.00	3.13	−0.13	−1.05	0.2996	No
Communication With Peers	2.74	2.87	−0.12	−0.94	0.3520	No
AgileM Efective-ness	3.49	3.60	−0.11	−0.78	0.4365	No
Belonging	2.92	3.03	−0.11	−0.84	0.4053	No
Lack Empl Engagement	3.20	3.30	−0.10	−0.60	0.5533	No
Boredom	2.43	2.33	0.09	0.62	0.5407	No
Knowledge Exchange	2.88	2.97	−0.08	−0.64	0.5268	No
ProjectMgmt Problems	2.92	3.00	−0.08	−0.63	0.5301	No
Company Appreciation	3.14	3.07	0.07	0.66	0.5083	No
Frustration	2.49	2.53	−0.04	−0.27	0.7881	No
Feedback	3.17	3.13	0.04	0.26	0.7944	No
Concentration	3.60	3.57	0.03	0.17	0.8638	No
Communication With Boss	2.88	2.90	−0.02	−0.16	0.8720	No
Knowledge About-Role	2.99	2.97	0.02	0.18	0.8597	No
Stress Level	2.25	2.27	−0.02	−0.11	0.9119	No
Time Management	3.78	3.80	−0.02	−0.12	0.9074	No
Shared Leadership	2.83	2.83	0.00	0.03	0.9733	No
Anger Reactions	2.47	2.47	0.00	0.01	0.9896	No

For individual contributors, high remote work (compared to low remote work) is associated with:

Positive Effects:

- Significantly higher Comfort (+0.59 points, p = 0.0030)
- Significantly higher Flexibility (+0.39 points, p = 0.0256)
- Significantly higher Autonomy (+0.39 points, p = 0.0115)
- Significantly higher Performance (+0.37 points, p = 0.0256)
- Significantly higher Confidence Level (+0.34 points, p = 0.0297)
- Significantly higher Creativity (+0.34 points, p = 0.0366)
- Significantly higher Problem Solving (+0.31 points, p = 0.0340)

- Significantly higher Leadership Problems (+0.30 points, $p = 0.0464$)

Negative Effects:

- Significantly lower Conflict Relationships (−0.29 points, $p = 0.0490$)

A.3 Annex C: Role-Based Comparison Tables

A.3.1 Table C.1: Comprehensive comparison of managers vs. individual contributors across all performance, motivation, and stress metrics

Table 9: Role-Based Comparison of Remote Work Effects

Metric	Effect on Individual contributors (Role 0)	Effect on Leaders (Role 1)	Different Effect Pattern?
Leadership Problems	+0.30 ($p = 0.0464$)	−0.67 ($p = 0.0217$)	Opposite directions
Creativity	+0.34 ($p = 0.0366$)	+0.78 (not sig.)	Significant for Individual contributors only
Confidence Level	+0.34 ($p = 0.0297$)	+0.55 (not sig.)	Significant for Individual contributors only
Conflict Relationships	−0.29 ($p = 0.049$)	−0.47 (not sig.)	Significant for Individual contributors only
Comfort	+0.59 ($p = 0.003$)	+0.32 (not sig.)	Significant for Individual contributors only
Autonomy	+0.39 ($p = 0.0115$)	+0.31 (not sig.)	Significant for Individual contributors only
Performance	+0.37 ($p = 0.0256$)	+0.63 (not sig.)	Significant for Individual contributors only
ProjectMgmt Problems	−0.08 (not sig.)	−0.83 ($p = 0.002$)	Significant for Leaders only
Confidence ToAsk Questions	+0.17 (not sig.)	+1.16 ($p = 0.0024$)	Significant for Leaders only
Coordination Issues	+0.25 (not sig.)	−1.04 ($p = 0.0053$)	Significant for Leaders only
Tasks assignment	+0.24 (not sig.)	+0.68 ($p = 0.04$)	Significant for Leaders only
Company Appreciation	+0.07 (not sig.)	+0.48 ($p = 0.0421$)	Significant for Leaders only
Feedback	+0.04 (not sig.)	+1.31 ($p = 0.0$)	Significant for Leaders only
Lack Empl Engagement	−0.10 (not sig.)	−1.07 ($p = 0.003$)	Significant for Leaders only
Communication With Boss	−0.02 (not sig.)	+0.56 ($p = 0.0217$)	Significant for Leaders only
Frustration	−0.04 (not sig.)	−0.85 ($p = 0.0167$)	Significant for Leaders only
Trust in Superior	+0.14 (not sig.)	+1.14 ($p = 0.0027$)	Significant for Leaders only
Motivation	+0.30 (not sig.)	+1.30 ($p = 0.0$)	Significant for Leaders only

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Table 9 continued from previous page

Metric	Effect on Individual contributors (Role 0)	Effect on Leaders (Role 1)	Different Effect Pattern?
Personal Relationships	−0.20 (not sig.)	+0.78 ($p = 0.0035$)	Significant for Leaders only
Team Milestones	+0.18 (not sig.)	+1.34 ($p = 0.0005$)	Significant for Leaders only
Belonging	−0.11 (not sig.)	+1.02 ($p = 0.0104$)	Significant for Leaders only
Attention to Others	−0.17 (not sig.)	+1.10 ($p = 0.0037$)	Significant for Leaders only
Work Environment	+0.16 (not sig.)	+1.12 ($p = 0.0061$)	Significant for Leaders only
Team Productivity	+0.24 (not sig.)	+1.23 ($p = 0.0135$)	Significant for Leaders only
Knowledge Exchange	−0.08 (not sig.)	+0.65 ($p = 0.0062$)	Significant for Leaders only
Collab Betw TeamMembers	+0.22 (not sig.)	−0.88 ($p = 0.0001$)	Significant for Leaders only
Communication Issues	−0.13 (not sig.)	−0.94 ($p = 0.0003$)	Significant for Leaders only
Confidence Betw TeamMembers	+0.21 (not sig.)	−0.65 ($p = 0.0155$)	Significant for Leaders only
Task Pressure	−0.16 (not sig.)	−0.50 ($p = 0.0471$)	Significant for Leaders only
Shared Leadership	+0.00 (not sig.)	−1.00 ($p = 0.0$)	Significant for Leaders only
Team Influence	−0.15 (not sig.)	+1.14 ($p = 0.0009$)	Significant for Leaders only
Problem Solving	+0.31 ($p = 0.034$)	+1.30 ($p = 0.0013$)	Stronger effect for Leaders
Flexibility	+0.39 ($p = 0.0256$)	+0.34 ($p = 0.0463$)	Similar effect for both

*Statistically significant at $p < 0.05$

These findings reveal an interesting pattern: the effects of remote work are quite different between leadership roles. Leaders (Role 1) experience far more pronounced benefits from remote work across multiple metrics. For leaders, remote work is associated with substantially higher team milestones achievement (+1.34 points), feedback (+1.31 points), motivation (+1.30 points), and problem-solving abilities (+1.30 points). They also report significantly lower issues related to coordination (−1.04 points), employee engagement (−1.07 points), and communication (−0.94 points).

Individual contributors (Role 0) still benefit from remote work, but their significant advantages are fewer and generally smaller in magnitude. They primarily gain in personal well-being areas: comfort (+0.59 points), autonomy (+0.39 points), performance (+0.37 points), creativity (+0.34 points), and confidence (+0.34 points).

The most notable finding is still the opposite effect on leadership problems between roles. Individual contributors report significantly more leadership problems in high remote environments (+0.30 points), while leaders experience significantly fewer leadership problems (−0.67 points). This suggests leaders perceive remote work as simplifying their leadership responsibilities, while individual contributors may feel less supported or guided in remote settings.

Communication and coordination dynamics show marked differences between the two groups. Leaders report significantly fewer coordination issues (−1.04 points) and communication issues (−0.94 points) in remote work, while individual contributors show no significant changes in these areas. This indicates leaders may be leveraging remote communication tools more effectively or experiencing fewer interruptions and meeting burdens.

Social connection metrics reveal another important contrast. Leaders experience significantly higher belonging (+1.02 points) and attention to others (+1.10 points) in remote work, whereas individual contributors show slightly negative effects for the same metrics. This finding suggests remote work may enhance leaders' social connections while potentially diminishing them for individual contributors.

Problem-solving is one of the few metrics that shows significant improvement for both groups, though with different magnitudes. While individual contributors report moderate improvements (+0.31 points), leaders experience much stronger benefits (+1.30 points). This suggests remote work may enhance cognitive performance across roles, but with leaders benefiting more substantially.

Flexibility is the only metric showing similar significant positive effects for both groups (+0.39 points for individual contributors and +0.34 points for leaders), indicating that increased workplace flexibility is a universal benefit of remote work regardless of role.

While leaders benefit more in most areas, individual contributors do show some unique advantages in remote settings. Individual contributors experience significantly higher comfort, autonomy, and creativity – areas where leaders see positive but non-significant improvements. This suggests remote work offers important well-being benefits for individual contributors even if they don't experience the same level of productivity and team coordination gains as leaders.

A.4 Annex D: Dependent Variable Definitions

The following definitions are based on the comprehensive factor analysis conducted by Castrillón & Acevedo (2024), who identified 83 distinct factors affecting software developer experiences through qualitative research with Colombian professionals:

A.4.1 Performance Metrics:

Performance: Overall effectiveness and quality of individual work output in achieving assigned tasks and objectives.

Team Productivity: Collective efficiency and output of the development team in delivering software products and meeting project goals.

Team Milestones: Achievement of key project checkpoints and deliverables within established timelines.

Concentration: Capacity to maintain sustained attention on specific tasks, ensuring cognitive resources are aligned with relevant objectives [definition adapted from Castrillón & Acevedo, 2024, factor #15].

Knowledge Exchange: Effective transfer of knowledge, skills, information and experiences within a team or organization, including documentation, training, and natural idea flow [definition adapted from Castrillón & Acevedo, 2024, factor #81].

Communication with Peers: Clear, effective and timely information exchange between team members at the same hierarchical level [definition adapted from Castrillón & Acevedo, 2024, factor #13].

Communication with Boss: Quality and effectiveness of information exchange between individual contributors and their direct supervisors or managers.

Employee Engagement: Level of emotional and professional involvement of employees toward their responsibilities and the organization [definition adapted from Castrillón & Acevedo, 2024, factor #14].

Leadership Problems: Challenges related to leadership capacity, guidance quality, and management effectiveness within teams [definition adapted from Castrillón & Acevedo, 2024, factor #55].

Agile Methodology Effectiveness: Success in implementing agile principles and practices for iterative and incremental software development [definition adapted from Castrillón & Acevedo, 2024, factor #56].

Task Assignment: Effectiveness in distributing responsibilities and tasks among team members [definition adapted from Castrillón & Acevedo, 2024, factor #32].

Confidence Between Team Members: Level of trust in the reliability, integrity and competencies of team members [definition adapted from Castrillón & Acevedo, 2024, factor #16].

Collaboration Between Team Members: Process of working jointly to achieve common objectives, integrating skills, knowledge and resources [definition adapted from Castrillón & Acevedo, 2024, factor #12].

Qualified Personnel: Availability of software developers with the necessary competencies, skills and experience to effectively fulfill job responsibilities [definition adapted from Castrillón & Acevedo, 2024, factor #63].

Project Management Problems: Challenges in planning, organizing, and supervising tasks, resources and teams for effective project execution.

Confidence to Ask Questions: Comfort level and psychological safety in seeking clarification, help, or information from colleagues and supervisors.

Knowledge About Role: Clear understanding of job responsibilities, expectations, and required competencies for effective performance.

Shared Leadership: Distributed leadership approach where team members share decision-making authority and leadership responsibilities.

A.4.2 Stress Metrics:

Stress Level: Physical, emotional and mental response to work demands perceived as excessive or challenging [definition adapted from Castrillón & Acevedo, 2024, factor #36].

Frustration: Emotional response to obstacles, setbacks, or unmet expectations in the work environment.

Task Pressure: Perceived burden due to work demands, tight deadlines, and organizational expectations [definition adapted from Castrillón & Acevedo, 2024, factor #66].

Coordination Issues: Challenges in planning, organizing and supervising tasks, resources and teams, including conflict resolution and decision-making [definition adapted from Castrillón & Acevedo, 2024, factor #20].

Confidence Level: Self-assurance and belief in one's abilities to successfully complete work tasks and responsibilities.

Flexibility: Capacity to offer options and adapt to changing circumstances while balancing priorities [definition adapted from Castrillón & Acevedo, 2024, factor #45].

Communication Issues: Problems in clear, effective and timely information exchange that facilitate mutual understanding [definition adapted from Castrillón & Acevedo, 2024, factor #13].

Team Trust: Belief in the reliability, integrity and competencies of team members and collaborative relationships [definition adapted from Castrillón & Acevedo, 2024, factor #16].

Time Management: Ability to effectively organize and control time spent on work activities to maximize productivity.

Amount of Hours Worked: Total time dedicated to work activities during a day or week [definition adapted from Castrillón & Acevedo, 2024, factor #54].

Anger Reactions: Emotional responses involving irritation, frustration, or hostility in work-related situations.

A.4.3 Motivation Metrics:

Motivation: Internal drive and enthusiasm to engage in work activities and achieve professional objectives [definition adapted from Castrillón & Acevedo, 2024, factor #61].

Comfort: Sense of ease, satisfaction, and well-being in the work environment and with work arrangements.

Trust in Superior: Confidence in the reliability, integrity, and competencies of direct managers and organizational leadership.

Problem Solving: Capacity to identify, analyze, and resolve work-related challenges and obstacles effectively.

Work Enjoyment: Level of satisfaction, pleasure, and positive emotions derived from work activities and professional engagement.

Work Interest: Degree of enthusiasm, curiosity, and engagement with work tasks and professional responsibilities.

Company Appreciation: Recognition and valuation of employees' contributions and efforts by the organization [definition adapted from Castrillón & Acevedo, 2024, factor #70].

Work-Life Balance: Capacity to maintain harmonious relationships between work responsibilities and personal needs [definition adapted from Castrillón & Acevedo, 2024, factor #33].

Feedback: Process of providing specific, clear and constructive comments that help improve performance and align with organizational objectives [definition adapted from Castrillón & Acevedo, 2024, factor #72].

Boredom: Lack of interest, stimulation, or engagement with work tasks, often resulting from repetitive or unchallenging activities [definition adapted from Castrillón & Acevedo, 2024, factor #79].

Attention to Others: Consideration and responsiveness to colleagues' needs, perspectives, and contributions.

Job Loss Fear: Constant concern about employment stability and job security [definition adapted from Castrillón & Acevedo, 2024, factor #58].

Work Environment: Physical, cultural and interpersonal conditions in the workplace [definition adapted from Castrillón & Acevedo, 2024, factor #2].

Team Influence: Ability to participate in decision-making or influence organizational processes [definition adapted from Castrillón & Acevedo, 2024, factor #49].

Personal Relationships: Interpersonal bonds and connections between team members [definition adapted from Castrillón & Acevedo, 2024, factor #71].

Autonomy: Capacity to make independent decisions in work, including freedom to choose tools, approaches and methods that best fit project needs [definition adapted from Castrillón & Acevedo, 2024, factor #5].

Conflict Relationships: Tensions, disagreements, or problematic interactions between team members that affect work dynamics.

Belonging: Sense of connection, inclusion, and identification with team values, objectives, and organizational dynamics [definition adapted from Castrillón & Acevedo, 2024, factor #76].

Creativity: Capacity to generate new ideas, approaches or solutions that drive innovation in development processes [definition adapted from Castrillón & Acevedo, 2024, factor #22].

Note: Factor definitions marked with specific factor numbers reference the comprehensive 83-factor analysis conducted by Castrillón & Acevedo (2024). Definitions without specific factor references represent concepts that emerged from our role-based analysis but align with the theoretical framework established in their qualitative research.